5. (a) Arcturus is a red giant star, 37 light-years away from Earth.

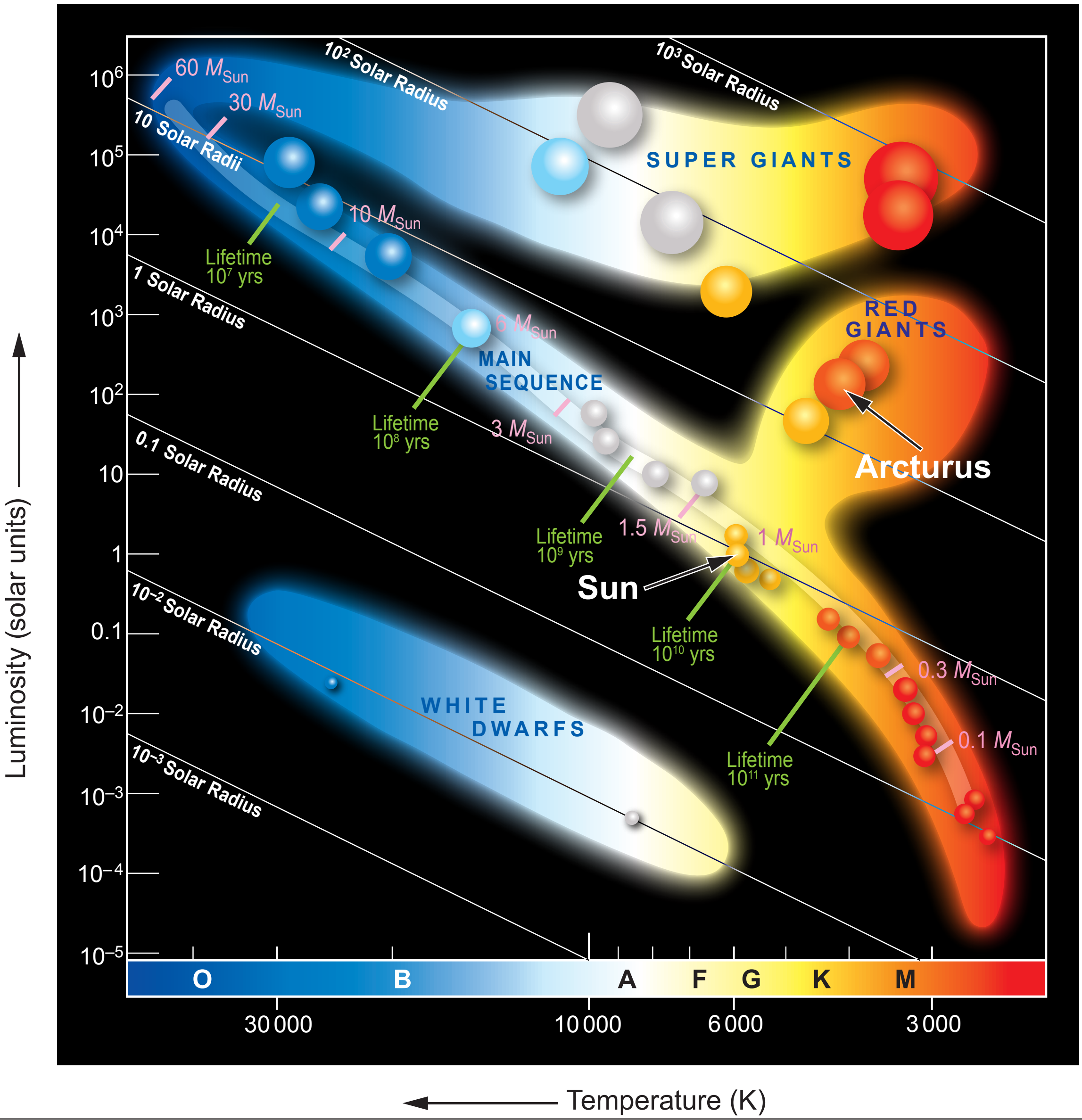
(i) State how long light from Arcturus takes to reach the Earth. [1]

time = ...

(ii) Use your answer to (i) and an equation from page 2 to determine the distance of Arcturus from Earth in metres.
(1 year = 31 600 000 s, speed of light, $c = 3 \times 10^8$ m/s) [3]

distance = ... m

(b) The Hertzsprung-Russell (HR diagram) below shows the properties of different stars.



Examiner only

(i) The Sun and Arcturus have different colours.
Use information from the diagram to compare their other properties. [2]

....

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....

(ii) The Sun is currently on the main sequence.
At the end of this stage of its life cycle it will become a red giant like Arcturus.
Explain, in terms of forces **and** fusion, why the Sun will become a red giant. [4]

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